***{06}* Team Charter**

Team Members	Team member 1: Haider_Shah (haider9676@gmail.com) Team member 2: Kamoru_Raji (Kamo4nice@gmail.com) Team member 3: Garvita_Aggarwal (garvita.agarwal15@gmail.com) Team member 4: Noor-e-hira_Noorani (noorehira.noorani@gmail.com) Team member 5: Rahul_Rock (rockrahul67392@gmail.com) Team member 6: Tosin_Ige (igetosin72@gmail.com) Team member 7: Victor_Agbayekhai (vijayagbaye@gmail.com)
Team Lead	Kamoru Olayemi Raji (Kamo4nice@gmail.com) Noor-e-hira_Noorani (noorehira.noorani@gmail.com)
Team Members Roles and Responsibilities	**Sponsor Company** – Excelerate **Company P.O.Cs** – Client, Role(s) in Company **Haider Ali Shah (haider9676@gmail.com) Project Manager:** Owns the project plan, timeline and task tracking; checks each deliverable against the evaluation criteria; supports the Team Lead on sponsor communications. **Kamoru Olayemi Raji (Kamo4nice@gmail.com) Team Lead 1:** **Noor-e-hira_Noorani (noorehira.noorani@gmail.com) Team Lead 2:** Single point of contact with the sponsor; represents the team on calls and via email; owns overall delivery quality and timeline. **Noor-e-hira_Noorani (noorehira.noorani@gmail.com)Project Scribe:** Captures meeting minutes, tracks action items, and distributes notes within 24 hours of each meeting; supports the Team Lead in drafting sponsor communications. **All other members:** Contribute to research, analysis, drafting, and review for their assigned workstream (database setup, dataset overview, SQL exploration and EDA, data quality assessment, Looker Studio visualization, QA) and complete individual deliverables by agreed internal deadlines.
Mission, Vision, Objectives, & Core Values	**Mission:** To deliver a rigorous, analysis-based report with clear visualizations, and a practical foundation the sponsor can act on. **Vision:** A collaborative, respectful team dynamic where every member's analysis strengthens the group's final report. **Core Values:** Integrity, Accountability, Discipline, Respect, Innovation.

Decision-Making Procedures	• Routine/day-to-day decisions (formatting, minor wording, query style, task sequencing) are made by the workstream owner without a full-team vote. • Significant decisions (scope interpretation, final analytical conclusions, data quality ratings, submission readiness) require majority agreement (5 of 8) reached during a scheduled team meeting. • If the team cannot reach majority agreement within one meeting cycle, the Team Lead makes the final call after hearing all viewpoints, and documents the rationale in the meeting minutes.
Communication Norms	• Primary channel: team WhatsApp/Slack group for daily coordination; email for anything involving the sponsor. • The Team Lead represents the team to the sponsor; all sponsor-facing emails are cc'd to the full team to minimize miscommunication. • Internal messages are acknowledged within 24 hours; sponsor emails are responded to within 48 hours. • Members listen to and respect each other's ideas and keep updates concise and actionable.
Meeting Cadence	• Full team check-in: every Monday, 5:00 PM IST (virtual), covering progress, blockers, and next-step assignments. • Sponsor sync: every Tuesday, 11:00 AM IST, led by the Team Lead with the Project Scribe taking notes. • Ad hoc workstream huddles as needed between scheduled meetings.
Conflict Resolution Guidelines	• Address disagreements directly and respectfully with the individual(s) involved before escalating. • Focus on the work product and evidence, not personal attribution, when resolving disagreements over analysis or findings. • If a conflict cannot be resolved between the parties, the Team Lead facilitates a structured discussion and, if needed, makes a final decision to keep the timeline on track.
Internal Checks, Balances & Reviews	• Each workstream owner submits a draft to the Quality Reviewer at least 24 hours before the internal deadline. • Weekly status updates are shared with the sponsor by Tuesday 5:00 PM; individual contributions are due internally by Tuesday 9:00 AM.

	• Deadlines this week: internal draft complete by [date to be confirmed]; final review [date to be confirmed]; submission by [date to be confirmed], 11:59 PM.

Data Visualization Associate Early Remote Internship
Team 06

Week 1 Deliverable: Data Setup & Initial EDA Report

Author: Team 06 | Environment: Supabase PostgreSQL & Looker Studio | Date: October 2026

1. Database Setup

To establish a robust and scalable analytics pipeline for the remote internship, a cloud-native PostgreSQL environment was deployed and integrated with Google Looker Studio. Below is the documentation of the setup process:

- **Connection Method:** Supabase Cloud PostgreSQL (Project Reference: fkdcbrljgljofzmahxfd).

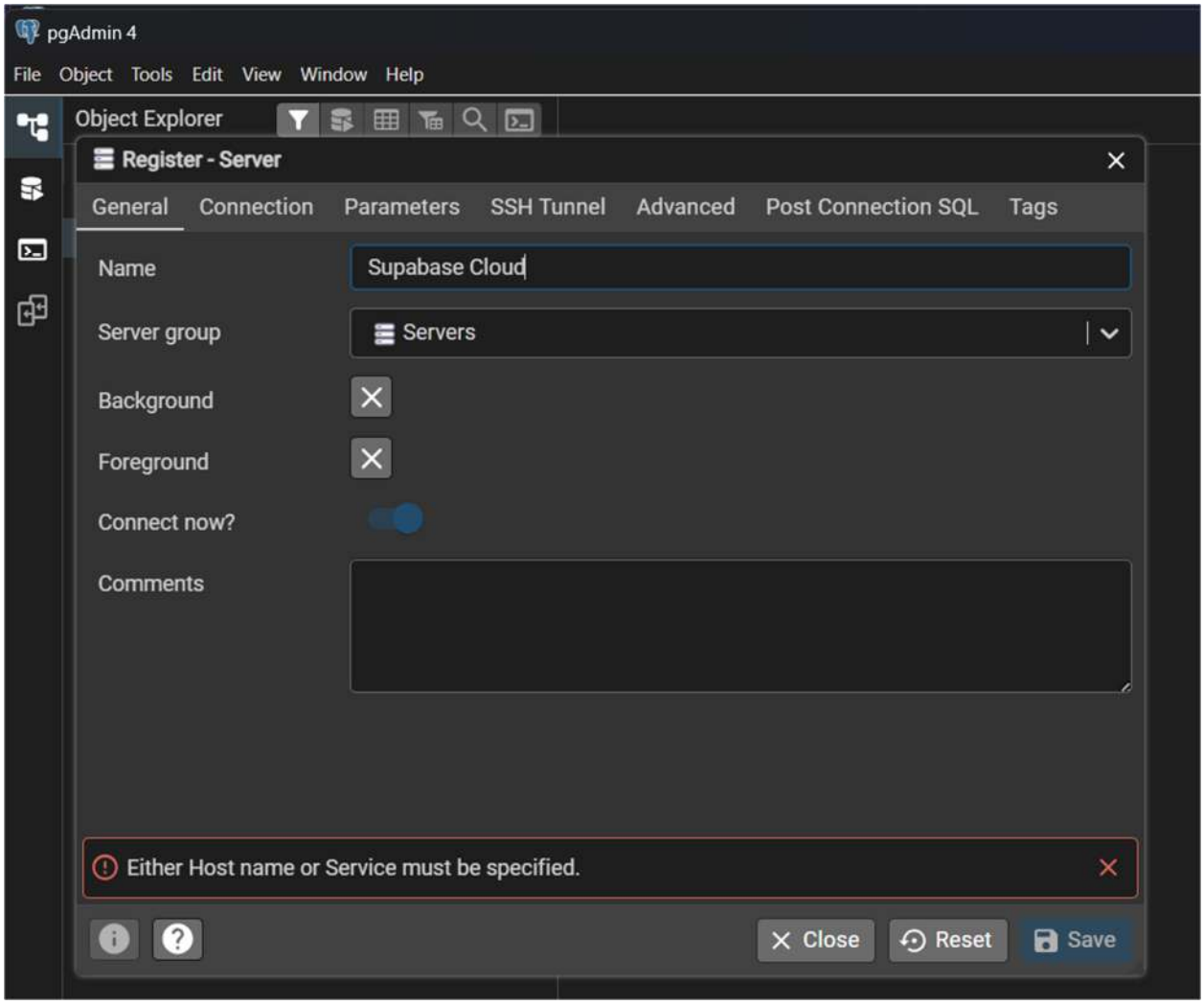


Figure 1: Supabase cloud connection to PostrgeSQL

- **Connectivity Architecture:** The connection was established via the Supabase Session Pooler (Supavisor) over port 6543 (Session mode) / 5432, ensuring reliable connectivity and bypassing local network restrictions.
- **Database Creation Process:** The database instance was provisioned with two primary schemas (`public.learner_interaction_data` and `public.opportunity_master_data`) created via DDL scripts in pgAdmin 4.

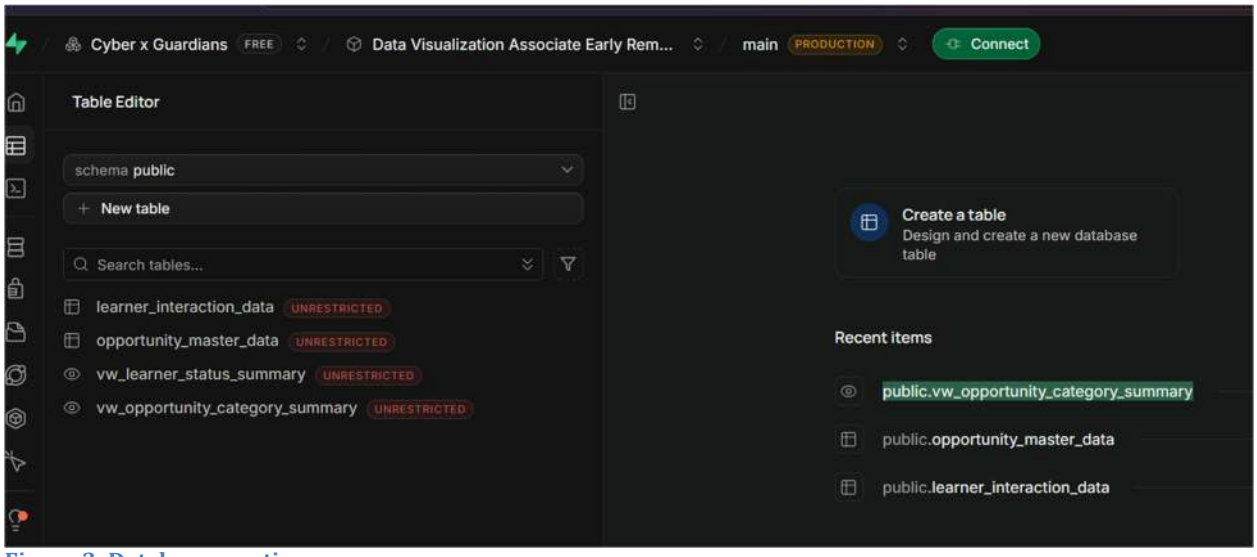


Figure 2: Database creation process

- **Dataset Upload Process:** CSV files provided in LMS were downloaded and the (`learner_data.csv` and `opportunity_data.csv`) were ingested into the staging tables using pgAdmin's import wizard with UTF-8 encoding and comma delimiters.
- **Confirmation of Successful Load:** Successful record counts (15,029 learner interaction rows and 5,733 opportunity master rows) were verified using SQL aggregation queries and confirmed within both pgAdmin and Looker Studio data source connections.

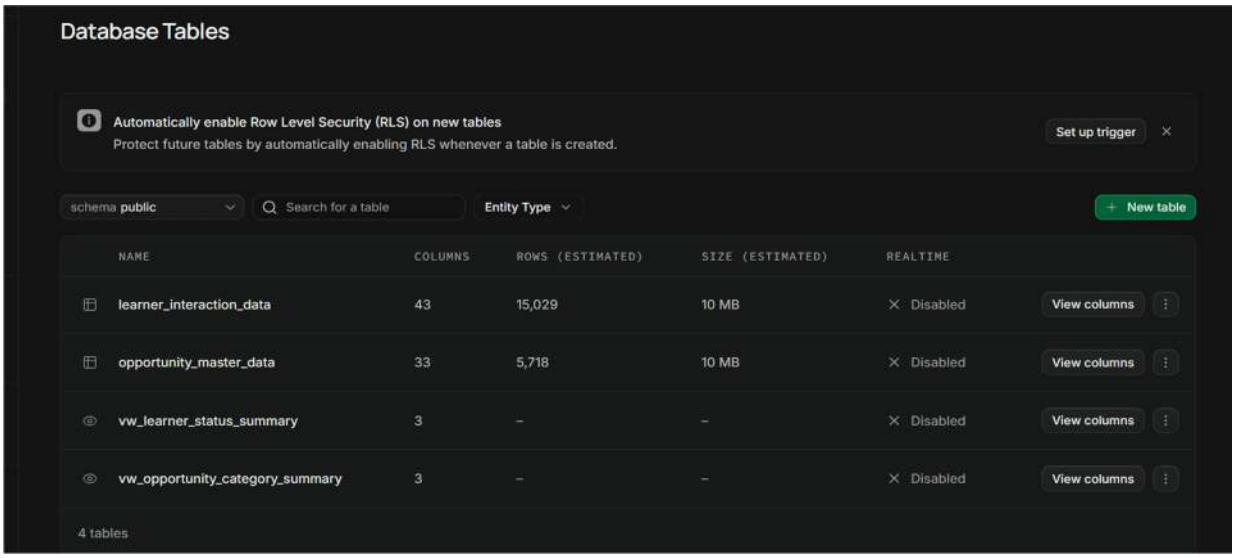


Figure 3: Confirmation of successful load

Data Visualization Associate Early Remote Internship

Team 06

2. Dataset Overview

The project analyzes two core datasets representing platform interactions and program catalogs:

2.1 Learner Interaction Dataset

- **Record Count & Structure:** 15,029 records spanning 43 comprehensive columns.
- **Key Fields:** `pk`, `sk`, `assigned_cohort`, `fee`, `status`, `accept_reject_date`, `amount_to_be_paid`, `transaction_id`, `payment_status`, `category`, `completion_date`, `created_at`, along with survey fields and timestamps.
- **Important Variables:** Tracks student engagement progression, payment tracking, and application statuses.
- **Initial Observations:** High volume of missing values in optional survey and open-ended feedback attributes (e.g., student status, custom essay questions), which is typical for multi-step onboarding forms.

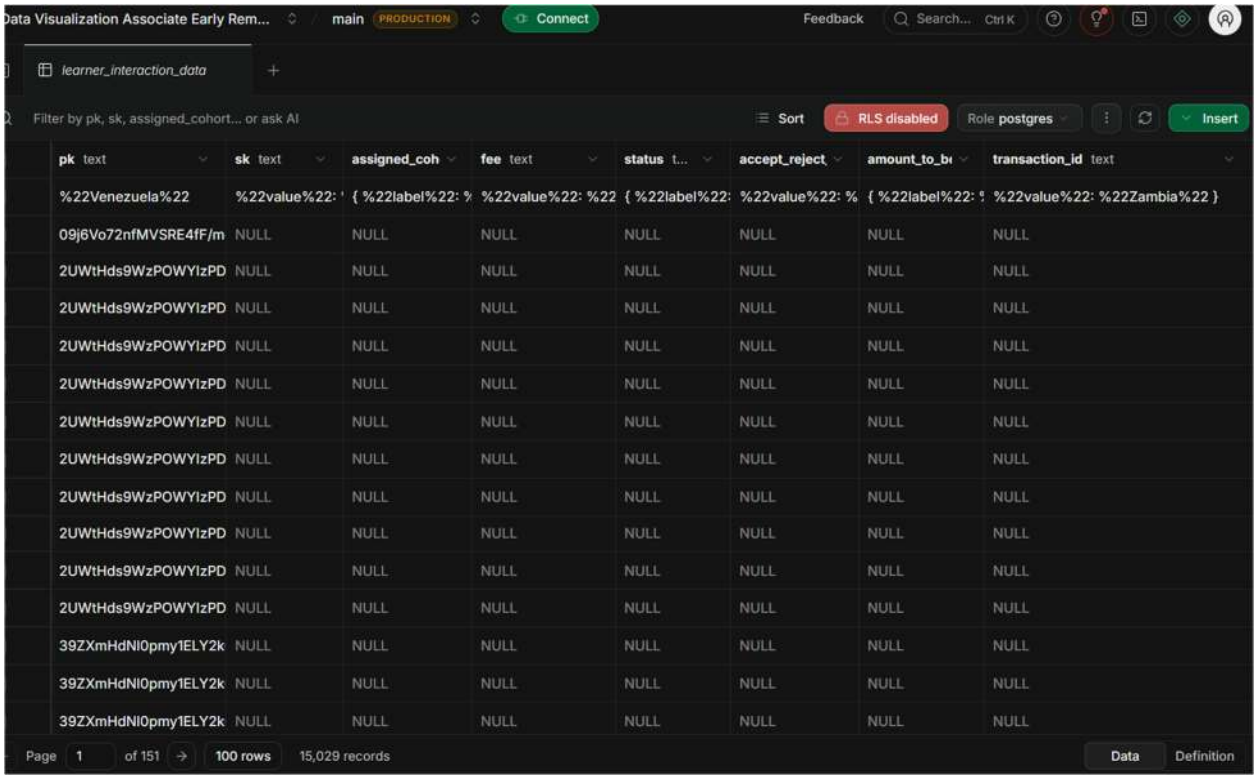


Figure 4: Learner interaction dataset

2.2 Opportunity Master Dataset

- **Record Count & Structure:** 5,733 records spanning 33 comprehensive columns.
- **Key Fields:** `pk`, `opportunity_id` (Primary Key), `category`, `fee`, `location`, `role`, `is_archived`, `created_at`, `duration`, `duration_type`, `name`, `summary`.
- **Important Variables:** Program classification, pricing/fees, geographical distribution, and role specifications.

- **Initial Observations:** Represents a master catalog of diverse learning tracks, competitions, internships, and career opportunities offered on the platform.

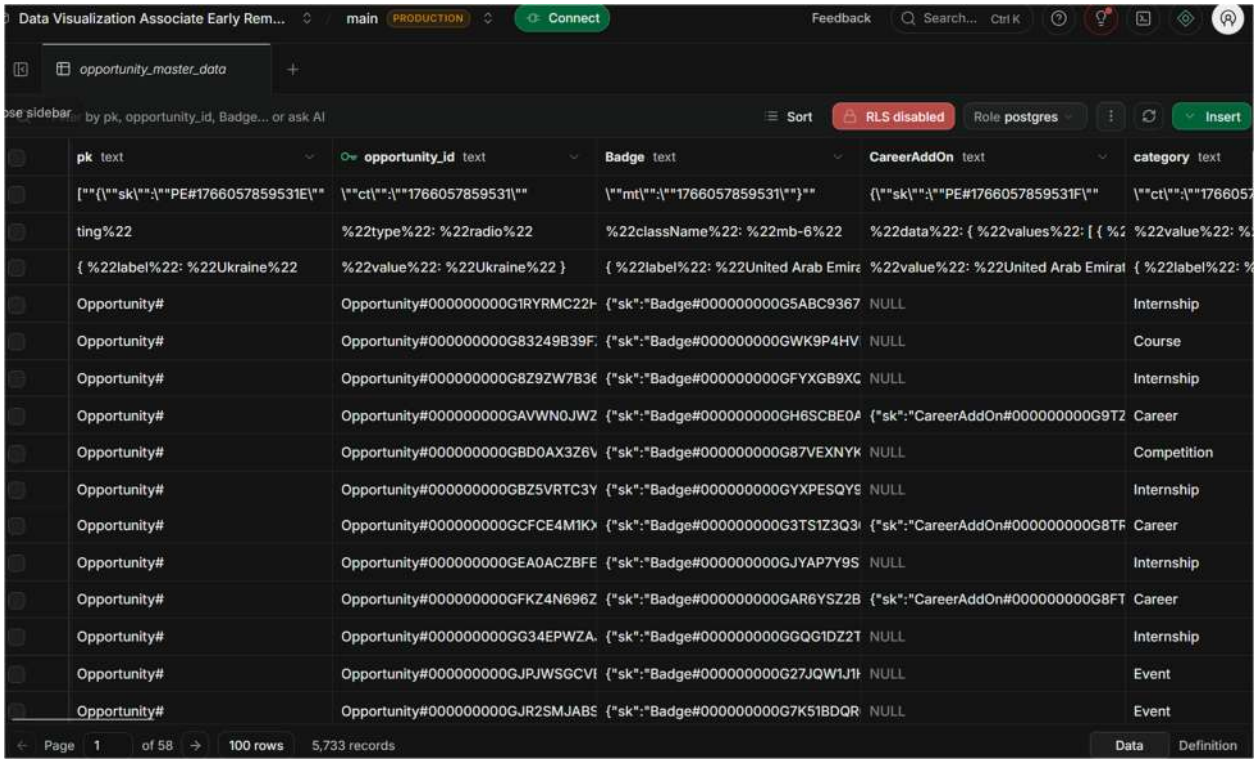


Figure 5: Oppertunity Master dataset

3. Exploratory Analysis

Initial exploratory data analysis (EDA) was conducted using SQL queries in PostgreSQL, followed by visualization in Google Looker Studio to uncover underlying distributions and operational trends.

3.1 SQL Queries Used for Exploration

```
-- 1. Opportunity Distribution by Category
SELECT category, COUNT(*) AS total_opportunities,
       ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) AS pct_share
FROM public.opportunity_master_data
GROUP BY category ORDER BY total_opportunities DESC;

-- 2. Learner Interaction Status Summary
SELECT status, COUNT(*) AS total_events,
       ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) AS pct_share
FROM public.learner_interaction_data
GROUP BY status ORDER BY total_events DESC;
```

3.2 Key Findings & Visual Distributions

- **Opportunity Category Breakdown:** The master catalog is heavily weighted toward Experiential Learning tracks, led by Internships (1,591 opportunities / 27.8%), Career Add-ons (1,088 / 19.0%), Competitions (734 / 12.8%), Courses (632 / 11.0%), and Events (622 / 10.8%).
- **Learner Status Progression:** Learner interactions demonstrate active progression through milestones, with Code 1070 (Completed/Accepted) representing the highest volume (~3,335 events), followed by Code 1010 (Applied/Registered) with ~1,791 events.
- **Potential for Deeper Analysis:** Opportunities for deeper analysis include studying cohort retention rates, analyzing fee-based conversion funnels, and correlating completion dates withspecific geographic regions.

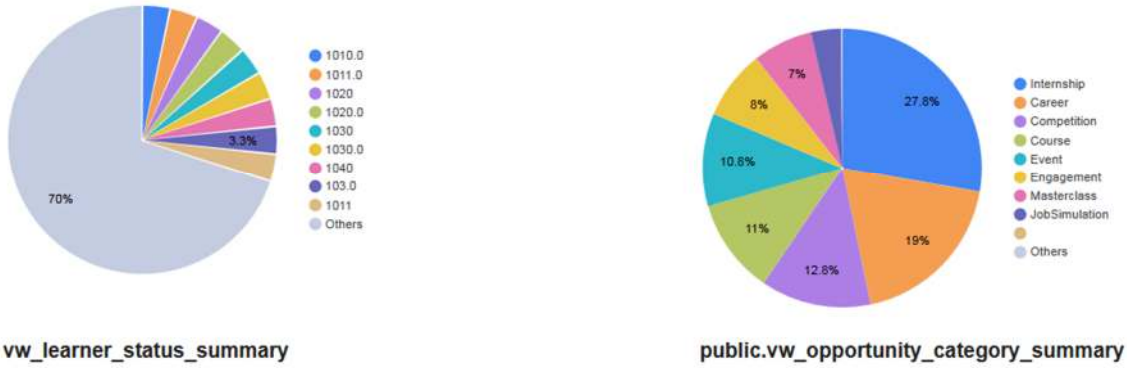


Figure 6: Key findings and Visual distributions

4. Data Quality Assessment

A rigorous data quality assessment revealed several structural and data hygiene challenges that must be addressed in subsequent pipeline transformations:

- **Missing Values:** Optional survey fields, feedback remarks, and transaction identifiers contain high null counts (e.g., `please_identify_your_student_status` has over 15,000 nulls; `transaction_id` has 14,831 nulls for unpaid or draft applications).
- **Duplicate Records:** Initial inspection identified 171 duplicate rows within the learner interaction staging table, requiring deduplication keys in analytical views.
- **Inconsistent Formatting:** Status codes appear in both numeric format (e.g., `1070.0`, `1010.0`) and string/integer formats without decimal points (e.g., `1070`, `1010`), leading to fragmentation in grouping operations.
- **Unexpected Values:** Presence of unparsed JSON/dictionary fragments or stray character strings within categorical columns in a small number of rows.
- **Impact on Future Analysis:** If unaddressed, mixed numeric/string status codes distort aggregation counts, and missing values in transactional fields require careful handling (treating nulls as unpaid or pending rather than dropping records) to prevent skewing conversion metrics.